

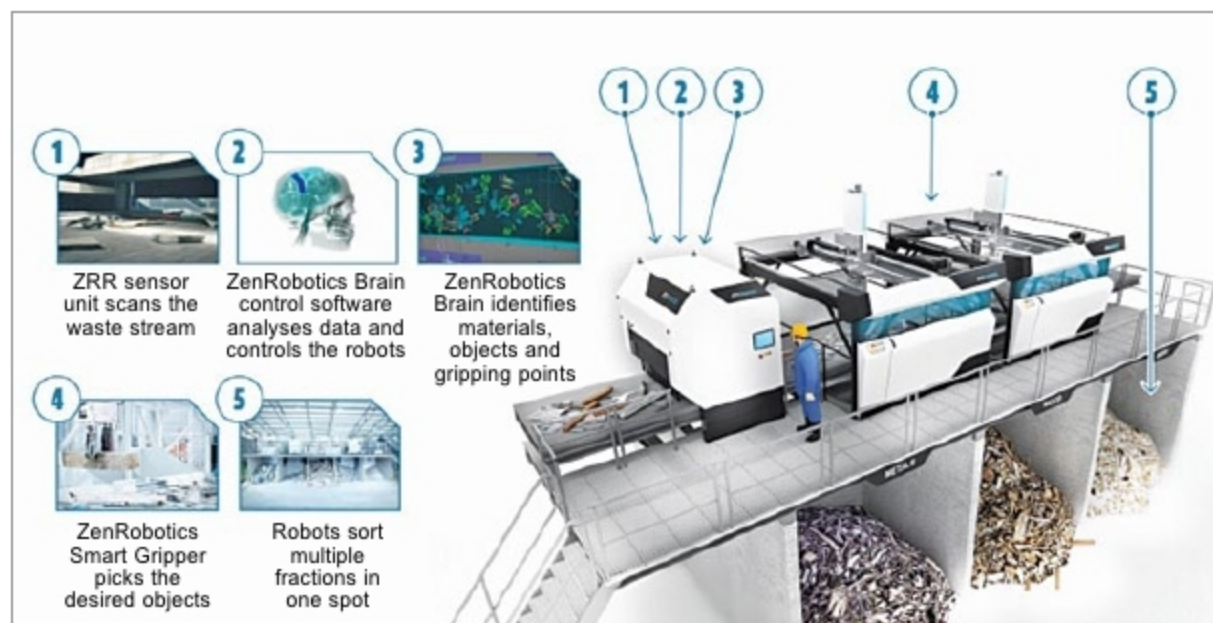


Fig. 3: Robotic waste sorting system by ZenRobotics (Credit: zenrobotics.com)

it is not that easy. Hence, product manufacturers should be part of e-waste management. This is where extended producer responsibility, or EPR, comes in. Manufacturers should try to be responsible through the entire lifecycle of the product, including taking back and disposing of the product.

In some countries, citizens have to pay for the waste produced by them. A number of firms have been fined for illegally dumping e-waste materials in public places.

Proper waste management, including e-waste, is required to prevent pollution and health issues. e-Waste management involves the following steps:

1. Collection
2. Segregation
3. Transportation
4. Processing
5. Disposal/incineration

Waste disposal in a landfill gives rise to land, air and water pollution. An alternative way is to burn/incinerate waste materials under controlled environments. Incineration is carried out both on small and large scales by industries using special incineration equipment under controlled conditions. A waste-to-energy plant is a waste management facility that combusts waste to produce energy.

e-Waste management involves both manual and large, intensive technical processes. e-Waste is first dismantled manually and then sent for further processing. For example, in a laptop, components like capacitors, memories, LCD screen and bat-

teries are first manually removed and separated. Then, the remaining part is sent for recycling through mechanical processing. The mechanical processing plant may include a crushing unit, shredders, and magnetic, eddy-current and air separators.

Burn the shit!

Once waste materials reach landfills, they all look the same and cannot be recycled properly. To get maximum benefit from waste materials, these first need to be sorted properly, either from trash bins or from disposal or recycling stations. Materials collected from trash bins are sent to disposal stations for recycling, incineration or to landfills.

Incineration not only helps get rid of waste, it converts waste into heat and electricity. So, it is advisable to burn materials that cannot be recycled and get the benefits of usable energy and heat while reducing pollution and health hazardous elements. However, this needs to be done under controlled environments.

AI in e-waste management

AI is the new saviour of e-waste management and recycling. AI, machine learning, image processing and robotics have a bright future in the waste management sector.

Two ways to sort waste materials before recycling are either dropping the waste into separate trash bins or letting the trash bins sort themselves through an automatic system.

If everything is dumped into one trash bin, sorting for recycling can

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